

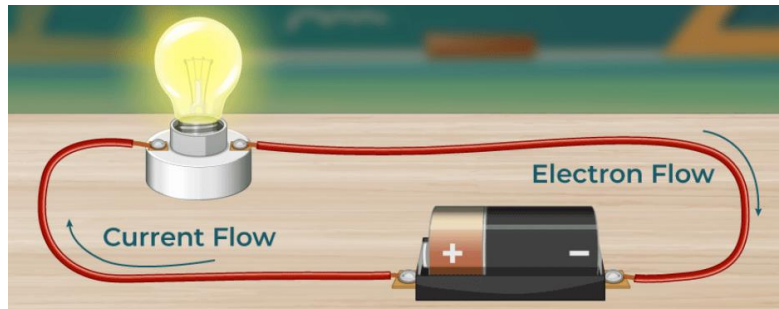
22PH202 – Electromagnetism and Modern Physics

Unit I - Electricity

1.5 Electric Current: Understanding the Flow of Electrical Charge

1.5.1 Introduction to Electric Current

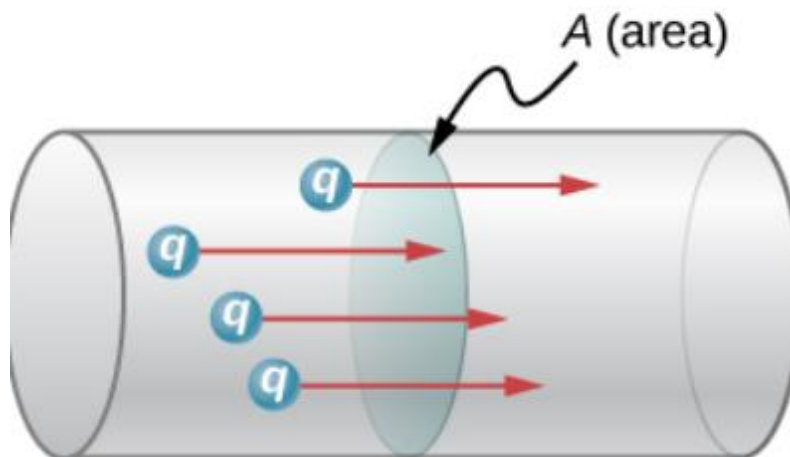
Electric current is a fundamental concept in physics and engineering, describing the flow of electric charge through a conductor. It plays a crucial role in various aspects of our daily lives, from powering electronic devices to enabling the functioning of complex electrical systems.



1.5.1.1 Definition of Electric Current:

Electric current is defined as the rate of flow of electric charge through a given area over time. It is typically denoted by the symbol 'I' and measured in Amperes (A).

Current = flow of charge



Mathematically, it is expressed as:

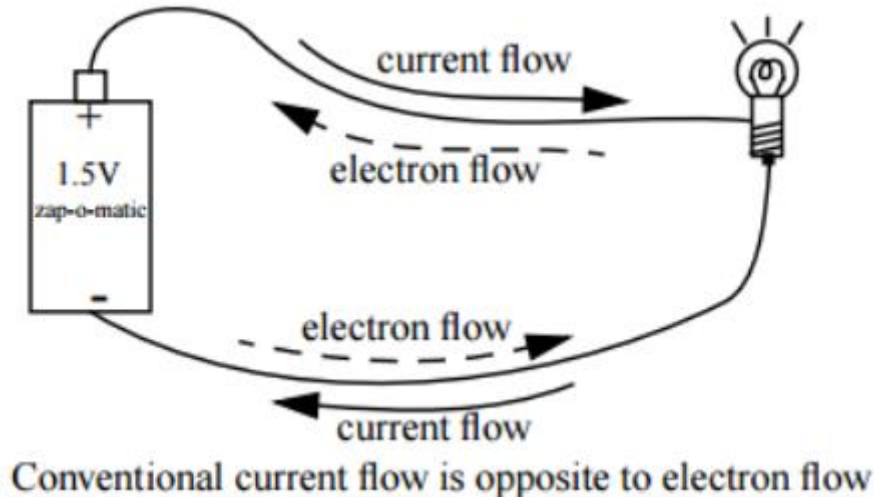
$$I = \frac{Q}{t}$$

Where:

- I = Electric current (Amperes)
- Q = Electric charge (Coulombs)
- t = Time (seconds)

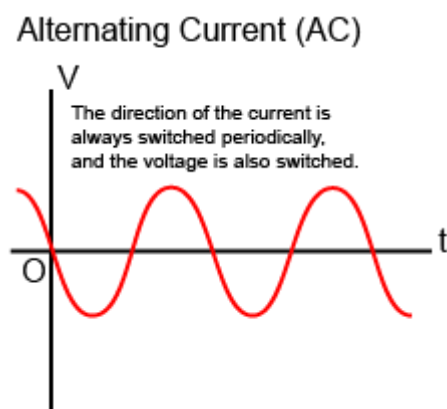
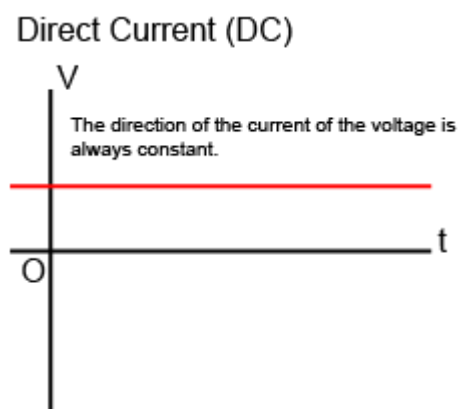
1.5.1.2 Direction of Current:

Conventionally, the direction of electric current is considered as the direction of flow of positive charges, although in reality, it involves the movement of electrons, which are negatively charged. This convention is known as conventional current flow.



1.5.1.3 Types of Electric Current:

Direct Current (DC): In DC, the electric charge flows continuously in one direction. Batteries and solar cells are common sources of DC.



Alternating Current (AC): In AC, the direction of the electric charge reverses periodically. It is the type of current supplied by most power grids and is used in many household appliances.

1.5.1.4 Factors Affecting Electric Current

Several factors influence the flow of electric current through a conductor:

1. Voltage:

Voltage, also known as electric potential difference, is the driving force that pushes electric charges through a circuit. It is measured in Volts (V). According to Ohm's law, the current flowing through a conductor is directly proportional to the voltage applied across it, given a constant resistance.

$$V=IR$$

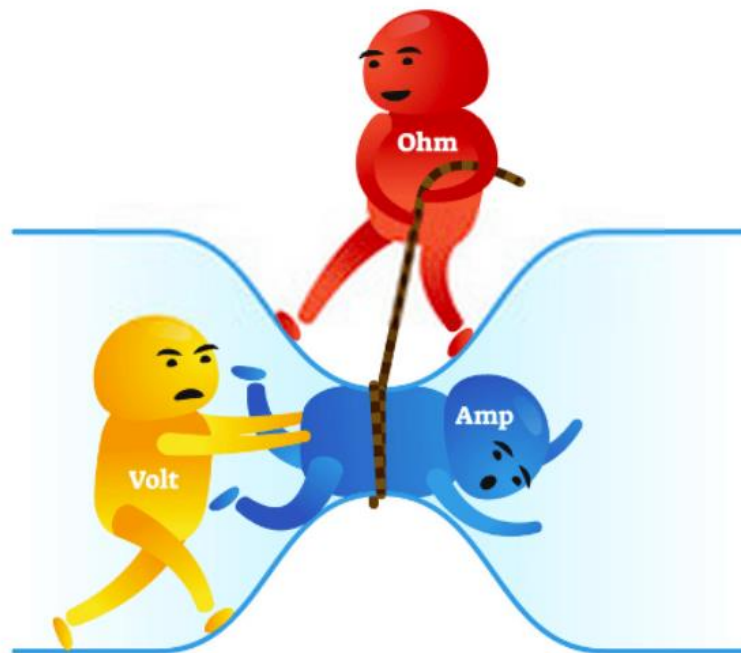
Where:

V = Voltage (Volts)

I = Electric current (Amperes)

R = Resistance (Ohms)

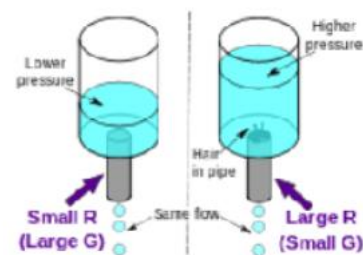
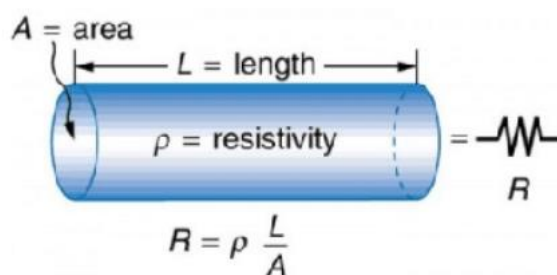
OHM'S LAW



2. Resistance:

Resistance is a property of a material that opposes the flow of electric current. It is measured in Ohms (Ω). Materials with high resistance impede the flow of current, while those with low resistance allow current to flow more easily.

What are the Factors Effecting the Resistivity of Electrical Materials?



3. Conductivity:

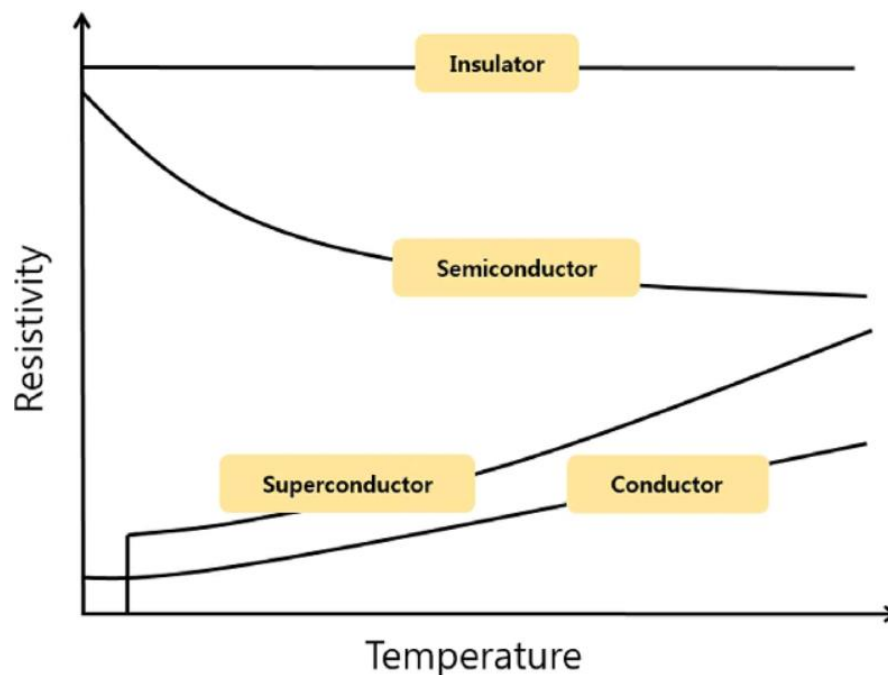
Conductivity is the inverse of resistance and describes how easily a material allows electric current to flow through it. Metals such as copper and aluminium are known for their high conductivity and are commonly used in electrical wiring.

Resistivity and Temperature Coefficient at 20 C

Material	Resistivity ρ (ohm m)		Temperature coefficient α per degree C	Conductivity σ $\times 10^7 / \Omega\text{m}$
Silver	1.59	$\times 10^{-8}$	.0038	6.29
Copper	1.68	$\times 10^{-8}$	.00386	5.95
Copper, annealed	1.72	$\times 10^{-8}$	.00393	5.81
Aluminum	2.65	$\times 10^{-8}$	.00429	3.77
Tungsten	5.6	$\times 10^{-8}$	.0045	1.79
Iron	9.71	$\times 10^{-8}$	.00651	1.03
Platinum	10.6	$\times 10^{-8}$	.003927	0.943

4. Temperature:

Temperature can affect the resistance of a material. In general, the resistance of most conductors increases with temperature, while the resistance of semiconductors may decrease with temperature.



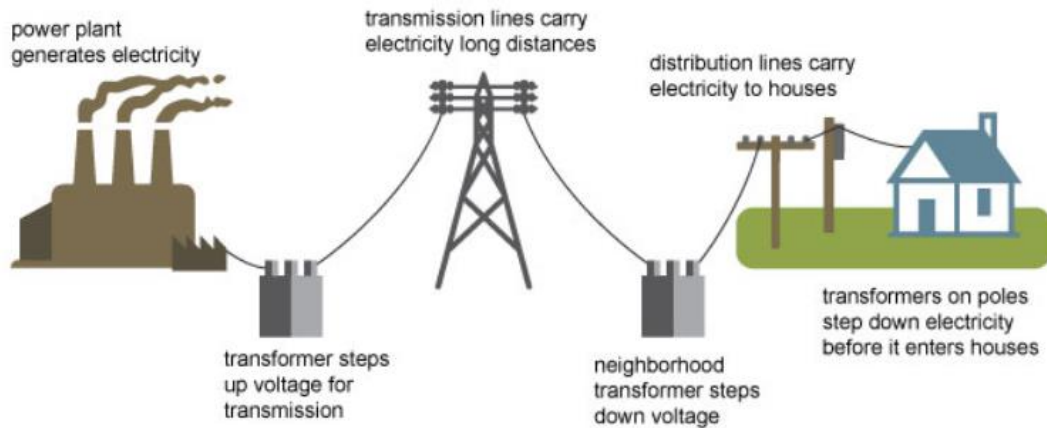
1.5.1.5 Applications of Electric Current

Electric current finds numerous applications across various fields:

1. Power Generation and Distribution:

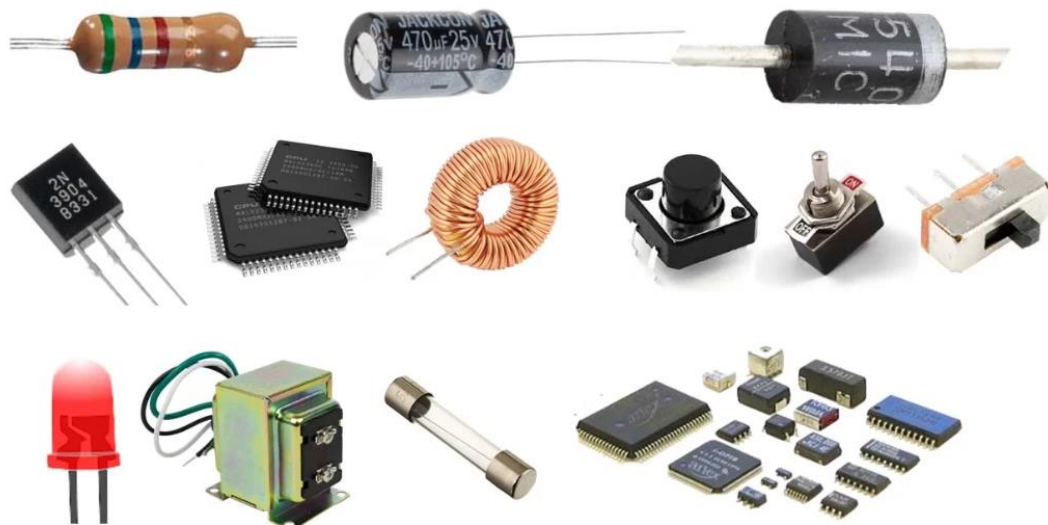
Electric current is used to generate power in power plants, which is then distributed through transmission lines to homes, businesses, and industries for various applications.

Electricity generation, transmission, and distribution



2. Electronics:

Electric current powers electronic devices such as smartphones, computers, televisions, and lighting systems. It enables the operation of integrated circuits, transistors, and other electronic components.



3. Heating and Lighting:

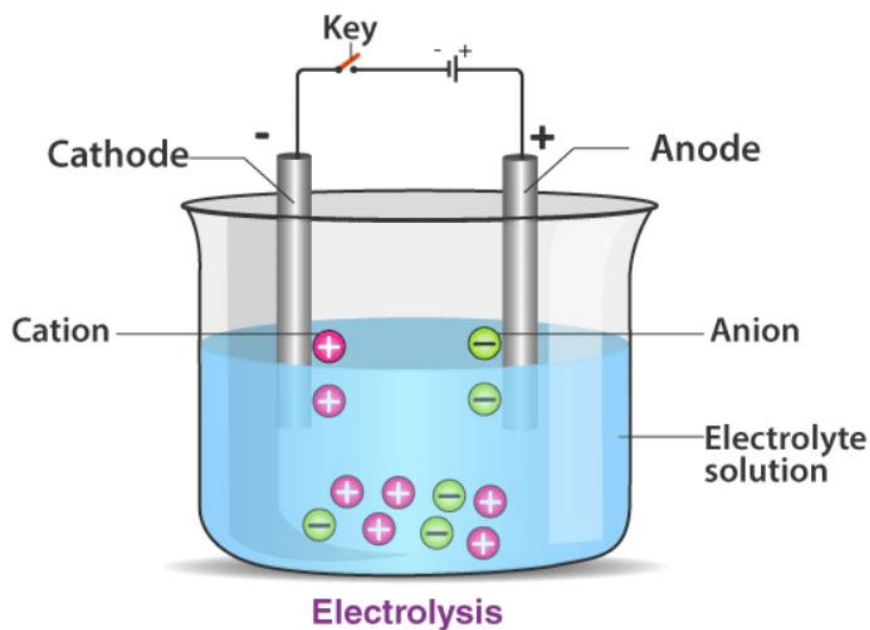
Electric current is used for heating in appliances like electric stoves, water heaters, and space heaters. It also powers lighting systems, including incandescent bulbs, fluorescent lamps, and LEDs.

Electric Heater



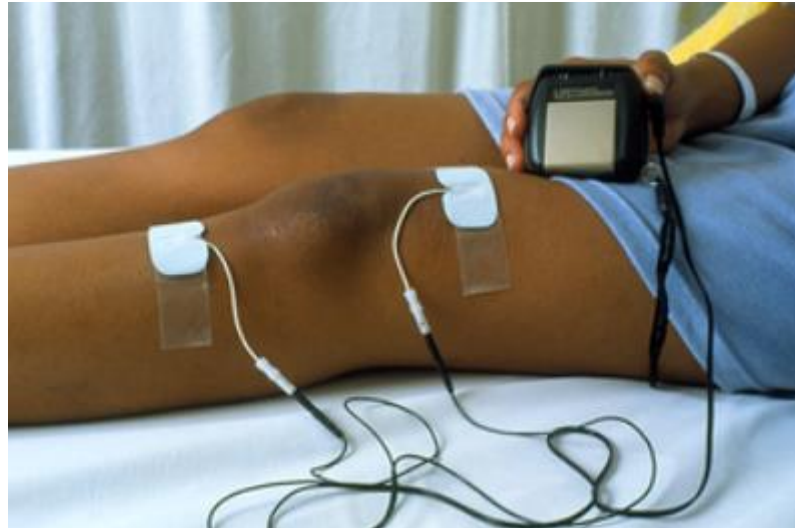
4. Electrolysis:

Electrolysis utilizes electric current to drive chemical reactions, such as the decomposition of water into hydrogen and oxygen gases or the electroplating of metals onto surfaces.



5. Medical Applications:

Electric current is used in various medical devices, including defibrillators, electrocardiograms (ECGs), and transcutaneous electrical nerve stimulation (TENS) units for pain relief.



Transcutaneous Electrical Nerve Stimulation (TENS)

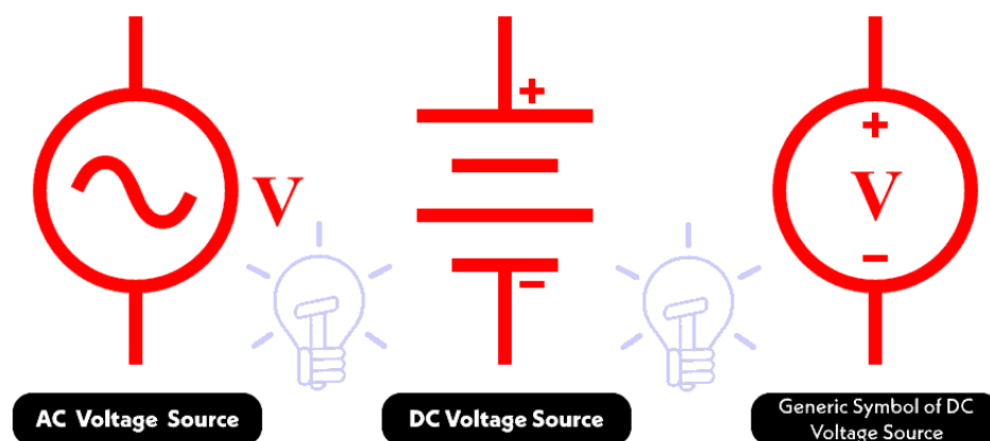
Electric current plays a pivotal role in modern society, powering a wide range of devices and systems that have become integral to our daily lives. Understanding its principles and applications is essential for anyone working with electricity or electronics.

1.5.2 Voltage Sources: Types, Characteristics, and Applications

1.5.2.1 Introduction to Voltage Sources

Voltage sources are essential components in electrical engineering and electronics, providing the necessary potential difference for the flow of electric current within circuits. This comprehensive guide explores the various types of voltage sources, their characteristics, and their wide-ranging applications.

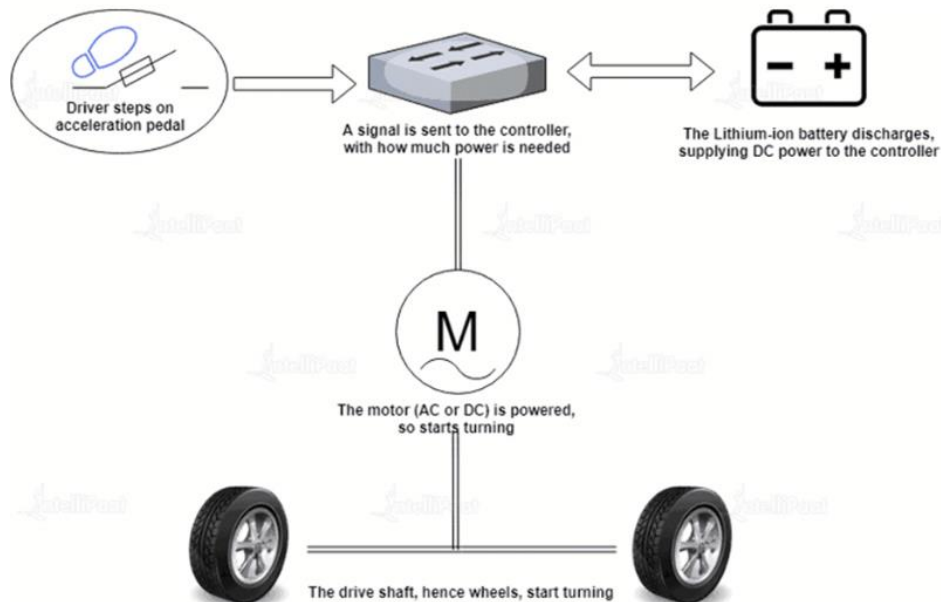
Voltage Source & Types of Voltage Sources



1.5.2.2 Definition of Voltage Source:

A voltage source is a device or component within a circuit that maintains a constant voltage difference across its terminals, facilitating the flow of electric charge. Voltage sources are

crucial for powering electronic devices, driving electric motors, and supplying energy to electrical systems.



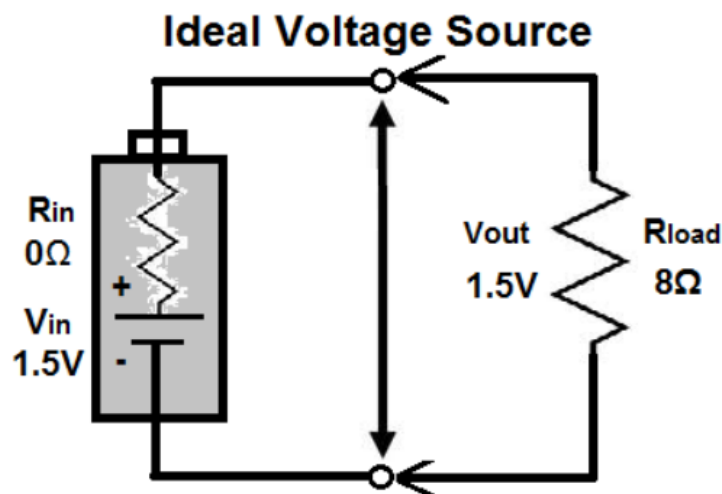
1.5.2.3 Importance of Voltage Sources:

Voltage sources serve as the driving force for electrical systems, enabling the transmission of power and the operation of electronic devices. Understanding the different types of voltage sources and their characteristics is essential for designing efficient circuits and systems.

1.5.2.4 Ideal and Practical Voltage Sources

1. Ideal Voltage Sources:

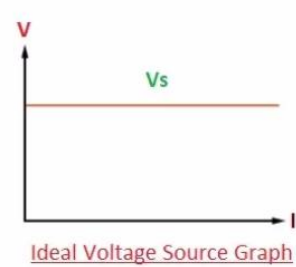
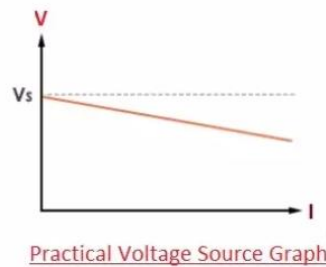
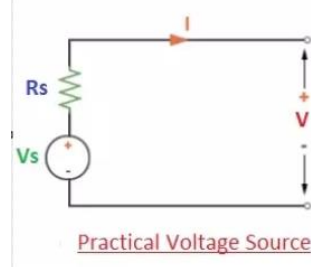
An ideal voltage source maintains a constant voltage across its terminals regardless of the connected load or the current flowing through it. In theoretical analysis, ideal voltage sources are represented as perfect sources of electrical energy, providing a fixed voltage under all conditions.



2. Practical Voltage Sources:

Practical voltage sources have internal resistance and may exhibit voltage variations under different load conditions. Real-world voltage sources such as batteries, power

supplies, and generators often have limitations and imperfections that affect their performance.



1.5.2.5 Direct Current (DC) Voltage Sources

1. Batteries:

Batteries are widely used DC voltage sources that convert chemical energy into electrical energy. They come in various types and sizes, including alkaline, lithium-ion, and lead-acid batteries, catering to diverse applications ranging from portable electronics to electric vehicles.



2. DC Power Supplies:

DC power supplies are electronic devices that convert alternating current (AC) from the mains supply into stable DC voltage suitable for powering electronic circuits and devices. They are commonly found in laboratories, industrial settings, and consumer electronics.

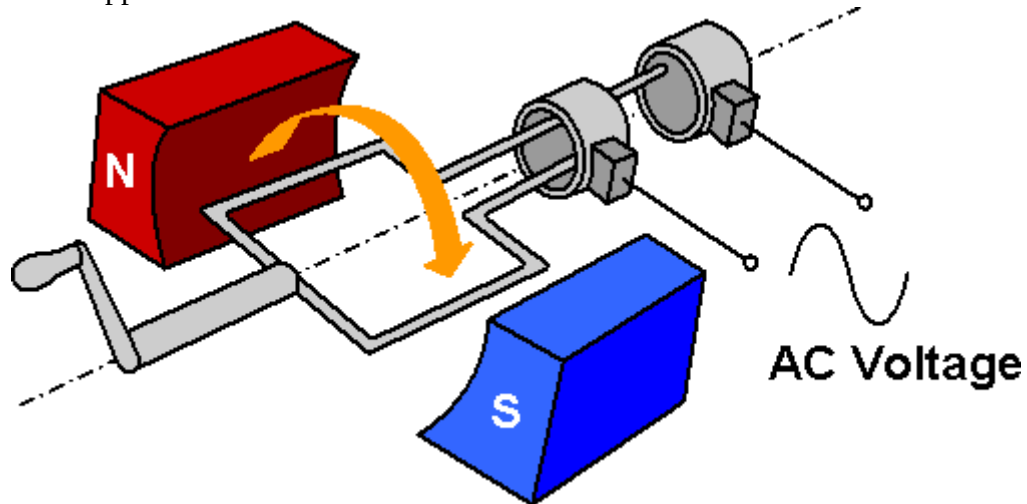
AC/DC Power Supply



1.5.2.6 Alternating Current (AC) Voltage Sources

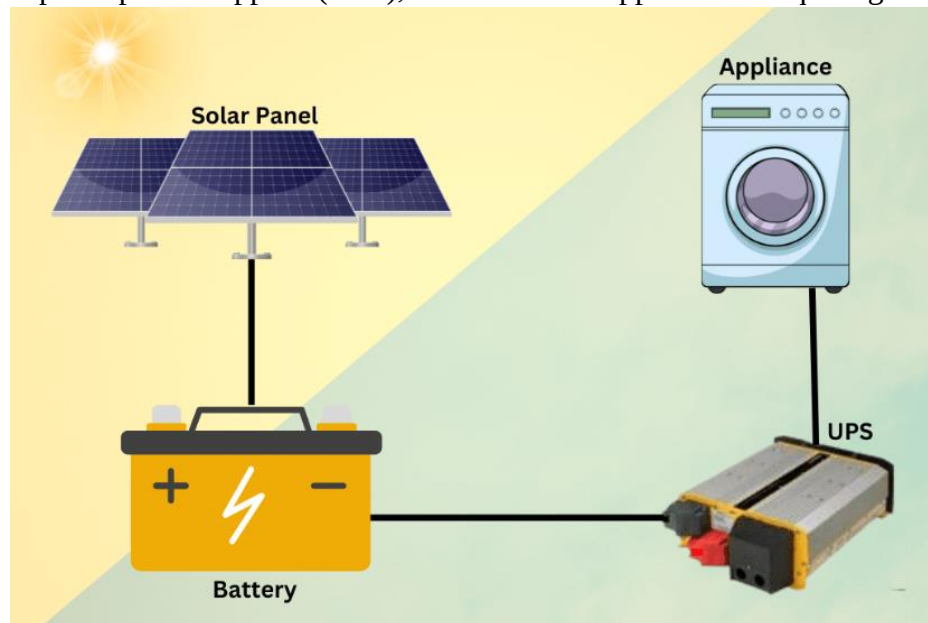
1. AC Generators:

AC generators, also known as alternators, produce alternating current by rotating a coil within a magnetic field. They are integral components of power plants and electrical generators, generating electricity for distribution across power grids and various industrial applications.



2. Inverters:

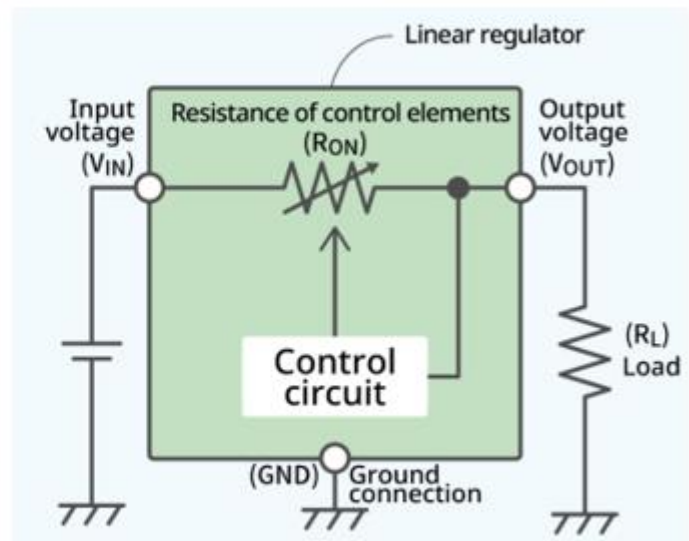
Inverters are electronic devices that convert DC voltage from sources such as batteries or solar panels into AC voltage. They play a crucial role in renewable energy systems, uninterruptible power supplies (UPS), and residential applications requiring AC power.



1.5.2.7 Characteristics of Voltage Sources

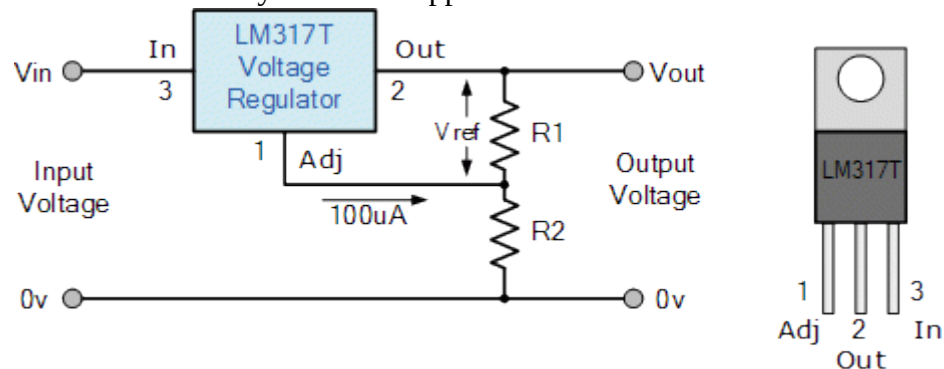
1. Voltage Regulation:

Voltage regulation refers to the ability of a voltage source to **maintain a constant output voltage despite changes in load or input conditions**. It is an important characteristic for ensuring stable operation of electronic circuits and devices.



2. Load Regulation:

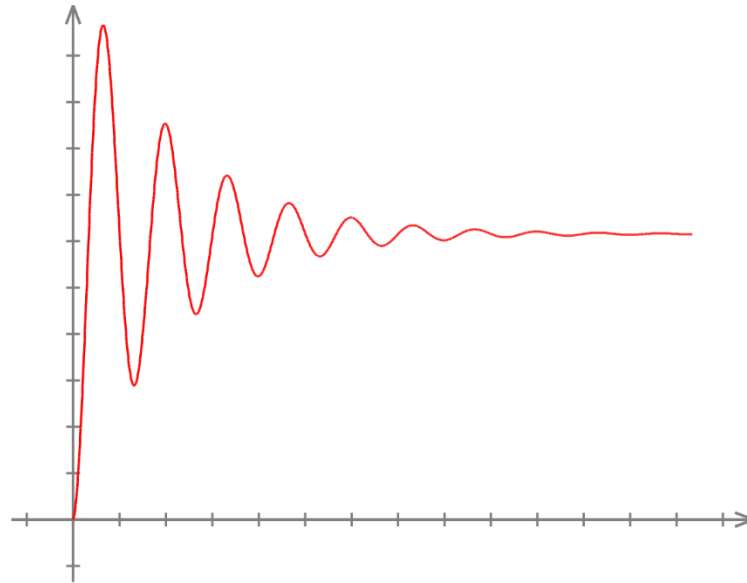
Load regulation measures the ability of a voltage source to **maintain a stable output voltage** as the load resistance varies. Low load regulation ensures consistent performance and reliability in diverse applications.



1.5.2.8 Transient Response and Efficiency

1. Transient Response:

Transient response describes how quickly a voltage source adjusts its output voltage following a change in load or input conditions. Fast transient response is essential for applications requiring rapid voltage changes, such as motor control and power electronics.



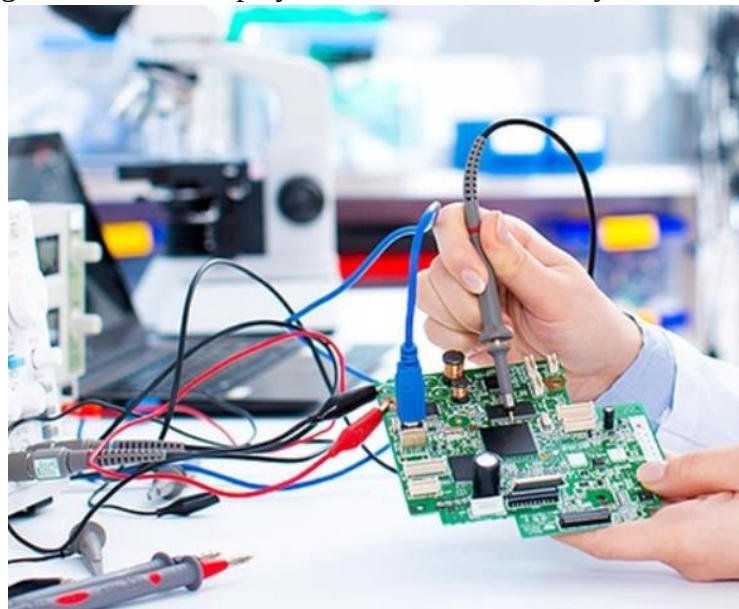
2. Efficiency:

Efficiency measures the effectiveness of a voltage source in converting input energy into useful output energy. High-efficiency voltage sources minimize energy losses and contribute to overall system performance and sustainability.

1.5.2.9 Applications of Voltage Sources in Electronics

1. Consumer Electronics:

Voltage sources power a wide range of consumer electronics, including smartphones, laptops, televisions, and home appliances. They provide the necessary voltage for powering integrated circuits, displays, and communication systems.



2. Industrial Electronics:

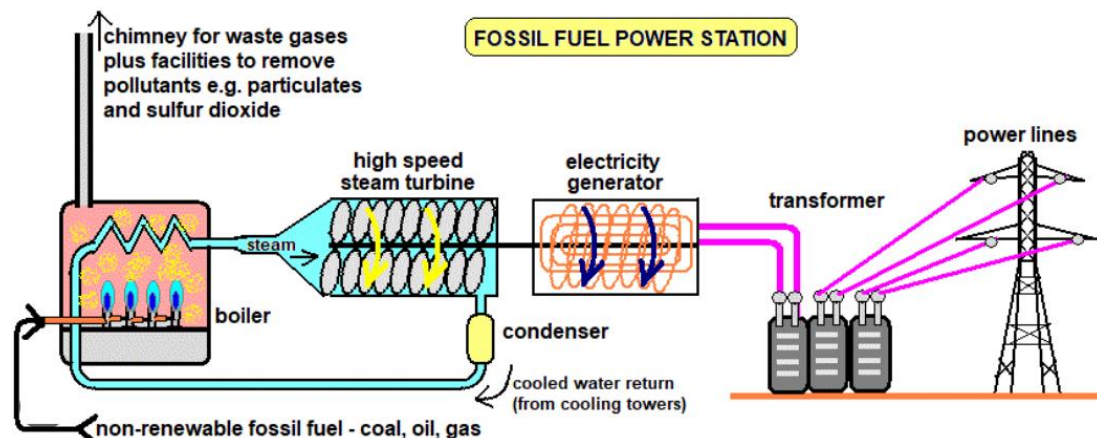
Industrial applications utilize voltage sources for motor control, process automation, and power distribution. Voltage sources drive industrial machinery, control systems, and manufacturing processes, contributing to productivity and efficiency.



1.5.2.10 Power Generation and Distribution

1. Power Plants:

Voltage sources in power plants generate electricity from various sources, including fossil fuels, nuclear energy, and renewable resources. AC generators produce alternating current for distribution across electrical grids.



2. Voltage Transformers:

Voltage transformers step up or step down the voltage levels to match the requirements of the electrical grid and end-users. They ensure efficient power transmission and distribution while maintaining voltage stability.



1.5.2.11 Renewable Energy Systems

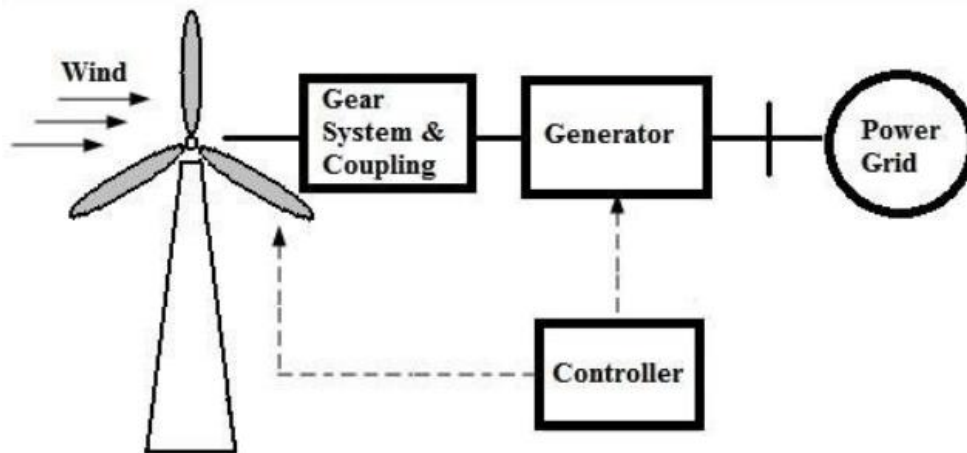
1. Solar Power Systems:

In solar power systems, voltage sources such as solar inverters convert DC voltage from solar panels into AC voltage for use in residential, commercial, and utility-scale applications. They enable the integration of solar energy into the electrical grid.



2. Wind Power Systems:

Wind turbines generate AC voltage from the mechanical energy of wind, which is then converted and synchronized with the grid voltage using power electronic converters. Wind power systems contribute to renewable energy generation and reduce dependence on fossil fuels.



1.5.2.12 Future Perspectives

Voltage sources are indispensable components in electrical engineering, electronics, and power systems, providing the necessary energy for a wide range of applications. As technology advances and energy demand grows, the development of efficient, reliable, and sustainable voltage sources will play a crucial role in shaping the future of electrical systems and renewable energy. Understanding the principles and characteristics of voltage sources is essential for engineers, researchers, and professionals working in diverse fields, contributing to innovation and progress in the global energy landscape.